



University of Asia Pacific

Department of Computer Science and Engineering

REPORT: 06

Course Code: CSE 402

Course Title: Operating System Lab

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Problem Statement:

Implement Non-Preemptive and Preemptive Priority CPU Scheduling for the following scenario, where No. of processes = 5, and the process information is:

Pid	AT	Priority	BT
P1	0	3	3
P2	1	2	4
P3	2	4	6
P4	3	6	4
P5	5	10	2

(A lower priority number indicates a higher priority.) Calculate WT, TAT, Average WT, and Average TAT for both scheduling methods. Also represent the process execution sequence for each.

Why is Priority Scheduling Used?

Priority Scheduling is a CPU scheduling algorithm in which every process is assigned a priority value, and the CPU is allocated to the process with the highest priority. In this implementation, a lower numeric value represents a higher priority.

In Non-Preemptive Priority Scheduling, once a process starts execution it runs to completion without interruption, even if a higher priority process arrives afterward. In Preemptive Priority Scheduling, a running process can be interrupted if a new process arrives with a higher priority (a lower priority number), and the CPU is immediately given to that new process. Preemptive priority scheduling generally responds faster to urgent processes, while non-preemptive scheduling avoids the overhead of frequent context switching.

Result: Non-Preemptive Priority Scheduling

Process	AT	Priority	BT	WT	TAT
P1	0	3	3	0	3
P2	1	2	4	2	6
P3	2	4	6	5	11
P4	3	6	4	10	14
P5	5	10	2	12	14

Average WT = 5.80 (Rounded: 6)

Average TAT = 9.60 (Rounded: 10)

Execution Sequence:

P1 → P2 → P3 → P4 → P5

Result: Preemptive Priority Scheduling

Process	AT	Priority	BT	WT	TAT
P1	0	3	3	4	7
P2	1	2	4	0	4
P3	2	4	6	5	11
P4	3	6	4	10	14
P5	5	10	2	12	14

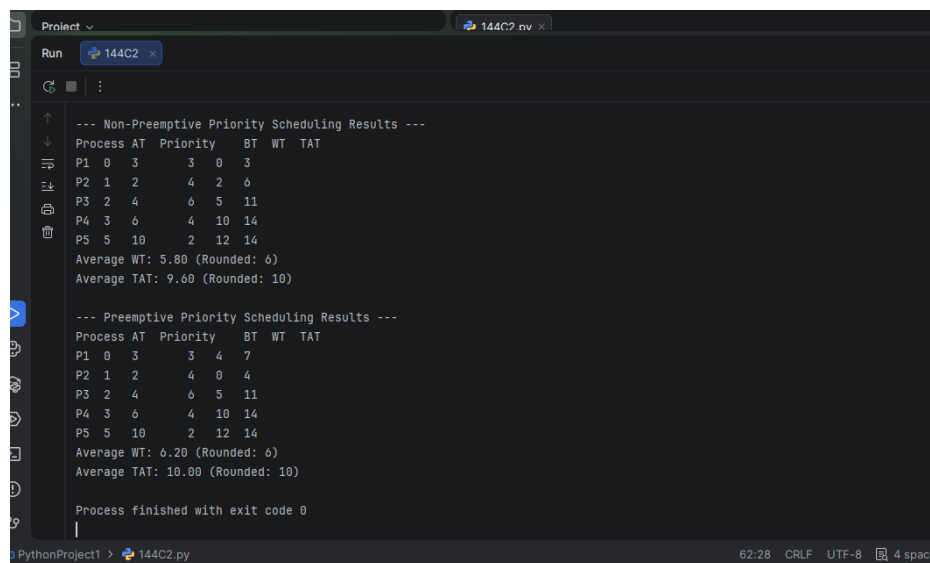
Average WT = 6.20 (Rounded: 6)

Average TAT = 10.00 (Rounded: 10)

Execution Sequence:

P1 → P2 → P1 → P3 → P4 → P5

Program Output:



```
--- Non-Preemptive Priority Scheduling Results ---
Process AT  Priority  BT  WT  TAT
P1 0 3 3 0 3
P2 1 2 4 2 6
P3 2 4 6 5 11
P4 3 6 4 10 14
P5 5 10 2 12 14
Average WT: 5.80 (Rounded: 6)
Average TAT: 9.60 (Rounded: 10)

--- Preemptive Priority Scheduling Results ---
Process AT  Priority  BT  WT  TAT
P1 0 3 3 4 7
P2 1 2 4 0 4
P3 2 4 6 5 11
P4 3 6 4 10 14
P5 5 10 2 12 14
Average WT: 6.20 (Rounded: 6)
Average TAT: 10.00 (Rounded: 10)

Process finished with exit code 0
```

Discussion:

In this experiment, both Non-Preemptive and Preemptive Priority CPU Scheduling were implemented using Python. In the non-preemptive version, the process with the highest priority (lowest priority number) among the arrived processes is selected and runs to completion. In the preemptive version, the CPU is re-evaluated at every time unit, and a running process is interrupted whenever a process with a higher priority arrives.

The main difficulty was correctly breaking ties when two processes had the same priority, which was resolved by selecting the process with the earlier arrival time. Another challenge was tracking the remaining burst time of each process in the preemptive version and computing the correct completion time whenever a process finished execution. The obtained Average WT and Average TAT for the non-preemptive method were 5.80 and 9.60 respectively, while for the preemptive method they were 6.20 and 10.00 respectively.

Conclusion:

The experiment successfully implemented both Non-Preemptive and Preemptive Priority CPU Scheduling. The program calculated the WT and TAT for all five processes under both scheduling schemes and determined their respective process execution sequences. This experiment helped in understanding how process priority, and the ability to preempt a running process, affect CPU scheduling performance and process waiting behavior.

Git Repository link:

<https://github.com/Rajash144/OS-CPU-Scheduling-Labs>